

Problem 1. (Importance of randomness and error) Prove that if a language $\mathcal{L}$ has an interactive proof with a deterministic verifier, then $\mathcal{L} \in \text{NP}$. Prove that if a language $\mathcal{L}$ has an interactive proof with zero soundness error, then $\mathcal{L} \in \text{NP}$.

Problem 2. (Sequential repetition) Suppose that $\mathcal{L}$ has an interactive proof (P, V) with perfect completeness and soundness error $1/2$. Let (P_t, V_t) be the t -fold sequential repetition of (P, V) : the new prover P_t and the new verifier V_t respectively simulate the old prover P and old verifier V for t times one after the other, each time with fresh randomness; V_t accepts if and only if V accepts in all t repetitions. Prove that (P_t, V_t) is an interactive proof for $\mathcal{L}$ with perfect completeness and soundness error 2^{-t} .

Problem 3. (Derandomised invertible matrices) Let $\mathbb{F}$ be a prime field such that $10n \leq |\mathbb{F}| \leq \text{poly}(n)$. Give interactive proofs for the language

$$\text{INV}_{\mathbb{F}} := \{M \in \mathbb{F}^{n \times n} : \exists A \in \mathbb{F}^{n \times n} \text{ s.t. } MA = I\}$$

with perfect completeness, soundness error $1/2$, where the verifier runs in time $\tilde{O}(n^2)$, and with each of the following additional properties:

- (a) $\tilde{O}(n)$ total communication, or
- (b) where the verifier uses $O(\log n)$ random bits.

(Fun challenge problem: can we achieve both simultaneously? I don't know!)

Problem 4. (Multilinear arithmetisation) Prove that if there exists a polynomial-time computable arithmetisation A of 3-CNFs (i.e., a mapping of boolean formulas to arithmetic circuits) such that for all $x_1, \dots, x_n \in \{0, 1\}$, $A(\phi)(x_1, \dots, x_n) = 0$ if and only if $\phi(x_1, \dots, x_n)$ is false, and $A(\phi)$ is multilinear for all ϕ , then $\text{coNP} \subseteq \text{BPP}$. You may assume that the underlying field $\mathbb{F}$ is sufficiently large.

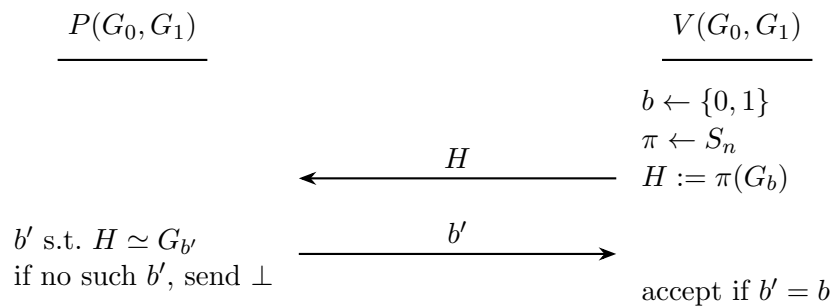
Prove that if, in addition, for all $x_1, \dots, x_n \in \{0, 1\}$, $A(\phi)(x_1, \dots, x_n) = 1$ if and only if $\phi(x_1, \dots, x_n)$ is true, and the characteristic of $\mathbb{F}$ is not 2, then there is a deterministic polynomial-time algorithm for $\#\text{SAT}$, the problem of counting satisfying assignments to a 3-CNF.

Problem 5. (Error reduction)

- (a) Let $S \subseteq \{0, 1\}^\ell$. Show that if $|S|/2^\ell \leq \frac{1}{3\ell}$, then for all $z_1, \dots, z_\ell \in \{0, 1\}^\ell$, $|\bigcup_i (S \oplus z_i)| \leq 2^\ell/3$, where $S \oplus z_i = \{s \oplus z_i : s \in S\}$. On the other hand, show that if $|S|/2^\ell \geq 2/3$, then there exist $z_1, \dots, z_\ell$ such that $\bigcup_i (S \oplus z_i) = \{0, 1\}^\ell$.
- (b) Denote by MA_1 the class of languages that have MA protocols with completeness 1. Show that $\text{BPP} \subseteq \text{MA}_1$.
- (c) Show that $\text{MA} = \text{MA}_1$.

(d) Show that $\text{MA} \subseteq \text{AM}$. (Hint: make the soundness error *very* small.)

Problem 6. (SZK for GNI) Recall the HVSZK protocol for graph non-isomorphism presented in class.



Prove that this protocol is (malicious verifier) statistical zero knowledge if and only if $\text{GI} \in \text{BPP}$.